ALL BRANCH (Hinglish)

General Aptitude

Quantitative Aptitude

DPP 02 Discussion Notes

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MCQ

The population of a city is increased by 30% and thus became 78000. What is the original population?

(A) 76000

(B) 64200

(C) 60000

(D) None

$1.3 \times x = 78{,}000$

$x = \frac{\cancel{78}^{6},000}{\cancel{1.3}_{1}}$

$= 60{,}000$

MCQ

A save 20% of his income. His income is increased by 20% and so he increased his expenditure by 30%. What is the percentage change in his savings?

(A) 20% fall

(B) 4% fall

(C) 20% rise

(D) 4% rise

I	S	Ex
100	20	80
120	16	104

$\frac{4}{20} \times 100 = 20\% \downarrow$

MCQ

In an election between two candidates, A and B, A secured 56% of the votes and won by 48000 votes. Find the total number of votes polled if 20% of the votes were declared invalid.

56% — 44%
A B

(A) 500000

(B) 400000

(C) 600000

(D) None

12% of D.V = 48,000

56 − 44 = 12%

$$D.V = \frac{48,000}{0.12} = 4,00,000$$

0.8 × T. Votes = 4,00,000

$$T.\ Votes = \frac{4,00,000}{0.8} = 5,00,000$$

MCQ

$\frac{300}{50} = 6/-$

A reduction of 10% in price of sugar enables a housewife to buy 5 kg more for Rs. 300/-. Find the reduced price per kg of sugar.

(A) 5/-

(B) 4.5/-

(C) 6/-

(D) None

$0.9 \times \frac{300}{x} = \boxed{\frac{300}{x+5}}$

$\frac{\overset{9}{\cancel{270}}}{x} = \frac{\overset{10}{\cancel{300}}}{x+5}$

$\Rightarrow 10x = 9x + 45$

$x = 45$

$\boxed{\frac{30}{5} = 6}$

MCQ

In a list of weights of candidates appearing for police selections, the weight of A is marked as 58 kg instead of 46.4 kg. Find the percentage of correction required.

wrong

correct

(A) 30

(B) 20

(C) 24

(D) None

$$\text{Correction\%} = \frac{\text{diff}}{\text{W. Value}} \times 100$$

$$= \frac{11.6}{58} \times 100 = \frac{1160}{58}$$

$$= 20\%$$

MCQ

In an examination, a student secures 40% and fails by 10 marks. If he scored 50%, he would pass by 15 marks. Find the minimum marks required to pass the exam.

10%

40% 50%

(A) 250

(B) 100

(C) 110

(D) 125

$$\begin{array}{r} 125 \\ -15 \\ \hline 110 \end{array}$$

25 marks

10% of Total Marks = 25

Total Marks = 250

MCQ

If 240 ltr of oil is poured into a tank, it is still 20% empty. How much more oil is to be poured to fill the tank?

(A) 300 ltr

(B) 60 ltr

(C) 120 ltr

(D) None

80% of Tank Capacity = 240

÷4 ÷4

20% of Tank Capacity = 60 lit

MCQ

A and B were hired for the same salary. A got two 40% hikes whereas B got a 90% hike. What is the percentage difference in the hikes they got?

(A) 16%

(B) 6%

(C) 10%

(D) 8%

A ⇒ $1.4 \times 1.4 = 1.96$

96% ↑

B ⇒ 90% ↑

$96 - 90 = 6\%$

MCQ

The population of a town increases at a rate of 10% for every year. If the present population is 12100, find the population two years ago.

(A) 11000

(B) 9800

(C) 10000

(D) 10120

$1.1 \times 1.1 \times x = 12100$

$1.21 \text{ of } x = 12100$

$x = \frac{12100}{1.21} = 10000$

MCQ

In an election 10 % of the votes were invalid. 40% of the votes were for A and the rest to B. B won with a majority of 243 votes, the total number of votes polled is

(A) 1250

(B) 1350

(C) 1215

(D) 1336

40% 20% 60%

A B

243

20% of D.V. = 243

$D.V = \frac{243}{0.2} = \frac{2430}{2}$

$= 1215$

90% of T.P.V = 1215

$T.P.V = \frac{1215}{0.9}$

$= \frac{12150}{9} = 1350$

Thank You!

GW Soldiers